

# A Multi-Task One-Stage Framework for Axial and Directional Object Detection

Is richer instance supervision compatible with higher observed box AP?

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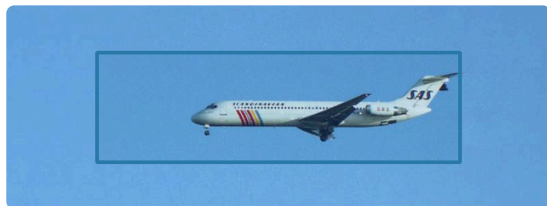
Mohamed Ben Hamdoun · Najib Gadi · Stéphane Gazut

Université Paris-Saclay · CEA, LIST · Smiths Detection, France

# Detection answers where and what — not how

Semantic orientation adds categorical axis and direction labels; it is not rotated-box angle regression.

## STANDARD DETECTION



box + class

+

## SEMANTIC ORIENTATION

O / D

AXIS

LR · FR · UD

DIRECTION

categorical labels

## RESEARCH QUESTION

Can richer supervision coexist with higher box AP under one shared recipe?

# Three contributions make the comparison inspectable

A private security X-ray need motivates the labels; public annotations and protocol metadata expose the comparison.

## 01 PUBLIC BENCHMARK

Curated COCO split with box, class, axis, direction and polygon annotations.

## 02 CONTROLLED STUDY

Four one-stage task heads share a backbone-neck and matched core optimization.

## 03 JOINT HEAD

Separate axis and direction branches join the detection and mask objectives.

## SCOPE TRANSFER

The surrogate transfers the supervision question — not operational X-ray appearance or deployment performance.

# A shared split with richer instance annotations

Every matched object has a box, class, categorical axis, categorical direction and polygon.

## IMAGES

5,281

4,753 train · 528 validation

## MATCHED OBJECTS

31,784

28,659 train · 3,125 validation

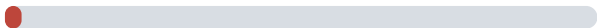
## LABEL STATES

2 + 3

axis + direction

## AXIS DISTRIBUTION

Diagonal  97.49%

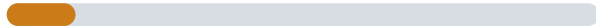
Orthogonal  2.51%

Severe class imbalance

## DIRECTION DISTRIBUTION

Front-Rear  44.85%

Left-Right  44.76%

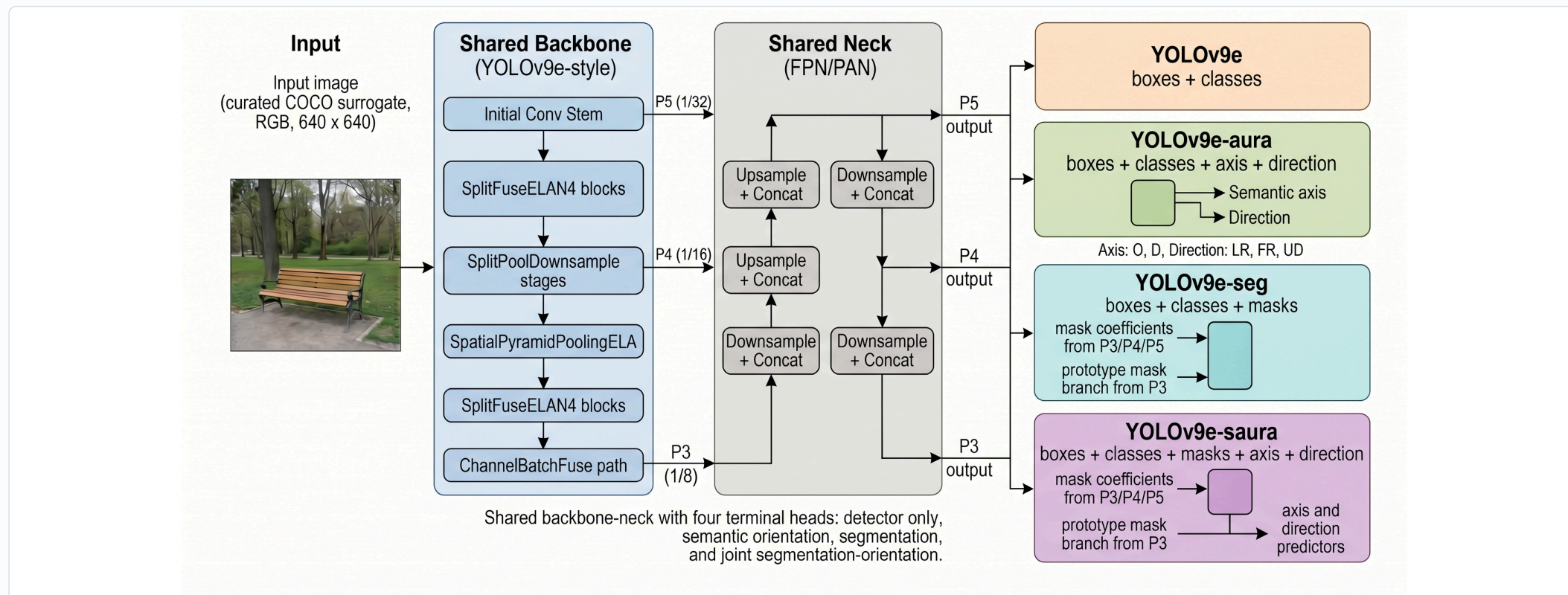
Up-Down  10.39%

## INTERPRETATION BOUNDARY

The split manifest retains 93 images without a matched object (not stated in the camera-ready body). Polygon completeness applies to matched instances; aggregate axis AP does not prove robustness on the rare Orthogonal state.

# Shared backbone-neck, four terminal heads

Published Fig. 2 · iCTrainer implementation · public COCO image, not operational X-ray · capacity not matched.



# Separate prediction branches, one weighted objective

The reported direction head is flat: LR, FR or UD directly, without hierarchical masking.

## ONE WEIGHTED OBJECTIVE · ONE JOINT STAGE

$$\begin{aligned}
 L = & \underbrace{\lambda_{\text{box}} L_{\text{box}} + \lambda_{\text{dfl}} L_{\text{dfl}} + \lambda_{\text{cls}} L_{\text{cls}}}_{\text{shared detection}} \\
 & + \underbrace{\lambda_{\text{axis}} L_{\text{axis}} + \lambda_{\text{dir}} L_{\text{dir}}}_{\text{Aura / Saura}} \\
 & + \underbrace{\lambda_{\text{seg}} L_{\text{seg}}}_{\text{Seg / Saura}}
 \end{aligned}$$

$$L_{\text{box}} = 0.45(1 - \text{CIoU}) + 0.55(1 - \text{NWD})$$

cls / axis / dir = BCE-logits · mask = area-normalized BCE (Dice off) · gains not retuned

## SHARED FEATURES, SEPARATE OUTPUTS

$$z_i = [r_i, c_i, a_i, d_i]$$

$$\hat{S}_i = \sum_k m_{i,k} P_k$$

from shared P3–P5 ·  $\hat{S}$  only on Seg / Saura

## ASSIGNMENT

$$M_i = s_{\text{cls},i}^{\alpha_{\text{obj}}} q_i^{\delta_{\text{loc}}}$$

$\alpha_{\text{obj}} = 1.0$ ,  $\delta_{\text{loc}} = 6.0$ ,  $q = \text{CIoU}$ ; axis / dir do not enter  $M$ ; they only backprop

**use\_aux\_in\_align=false**

# Matched core optimization, disclosed execution differences

All runs start from scratch on one H100; HSV jitter is active and geometric augmentation is disabled.

EPOCHS

**100**

INPUT

**640 × 640**

BATCH

**4**

OPTIMIZER

**SGD**

lr 0.01 · momentum 0.937

INITIALIZATION

**seed 0**

from scratch · AMP

HARDWARE

**1 × H100**

sequential runs

## MATCHED CORE

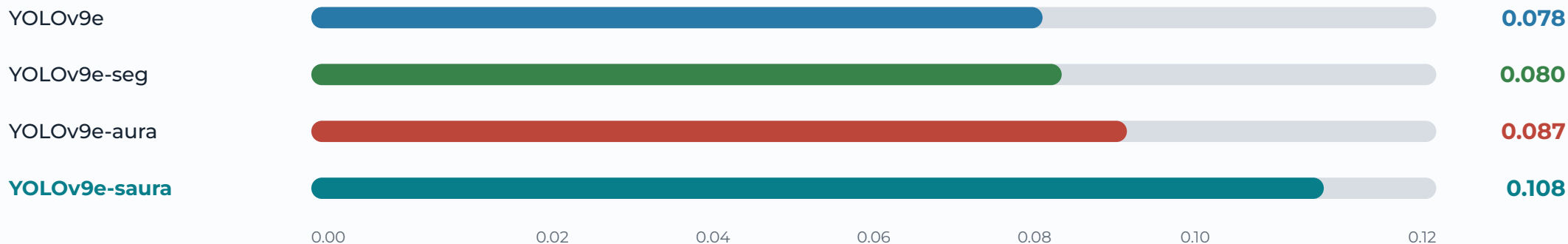
Same split, epochs, image size, batch, optimizer family, schedule, seed and precision mode.

## ARCHIVED DIFFERENCES

Seg validation cap: 300 detections versus 400. Saura data loader: 4 versus 8 workers, plus prefetch, pinning and persistence differences.

# The joint head reaches the highest observed box AP

Best-validation box mAP50–95. Bars are proportional from zero.



## PRIMARY OBSERVATION

Saura versus baseline: **+0.030 = +3.0 AP points**. Aura versus baseline: +0.009 = +0.9 AP points.

## INTERPRETATION

Compatible with higher observed AP under this recipe; not causal or statistically significant evidence.



# Higher metric-wise peaks come with a larger head

Best-validation mAP50–95 values; metric-wise peaks may come from different checkpoints.

MASK · SEG → SAURA

**0.062 → 0.087**

+0.025

AXIS · AURA → SAURA

**0.124 → 0.215**

+0.091

DIRECTION · AURA → SAURA

**0.120 → 0.130**

+0.010

PARAMETERS · BASELINE → SAURA

**58.15 → 71.08 M**

reported Table I estimates · +22%

COMPUTE · BASELINE → SAURA

**192.68 → 272.04**

reported GFLOPs · Table I estimates · +41%

## READING

The joint head has the highest displayed peaks, but the experiment does not isolate whether the difference comes from supervision, capacity, optimization interaction or their combination.

# One supported observation — and a clear boundary

## SUPPORTED HERE

- Richer supervision is compatible with higher observed validation box AP.
- The joint Saura head has the highest displayed metric-wise peaks.
- Public annotations, splits, protocol metadata and reconstruction scripts make the data protocol inspectable.

## NOT ESTABLISHED

- Causality or statistical significance: one seed and best validation.
- Compute-matched evidence: capacities and execution details differ.
- Operational X-ray transfer, deployment performance or state of the art.

## CORRECT CLAIM

Richer supervision is **compatible with higher observed AP** in this controlled study on a public surrogate — not proven to cause it.

# Three contributions, one bounded result

Public benchmark · controlled four-head comparison · joint axial and directional formulation.

## 01 PUBLIC DATA

Public annotations and split definitions for the surrogate.

## 02 FOUR HEADS

One shared trunk and matched core optimization.

## 03 JOINT LOSS

Detection, masks, axis and direction in one head.

# +3.0 AP

## Saura versus baseline box mAP50–95

public surrogate · single seed · best validation · capacity not matched

[github.com/mbenhamd/avss2026-curated-coco-benchmark](https://github.com/mbenhamd/avss2026-curated-coco-benchmark)

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## Questions?